

PROJECT REPORT

Campus Bus Tracking Application (CBTA)

GROUP MEMBERS

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1. System Requirements

1. User Authentication

- The system shall allow users to log in using a username and password.
- The system shall identify the user's role (Student, Driver, or Admin) after login.
- The system shall redirect users to their respective dashboards.

2. Student Management

- The system shall store student information including registration number, department, and semester.
- The system shall allow students to view their profile information.
- The system shall associate students with a bus through the STUDENT_BUS table.
- The system shall allow students to view their assigned bus details.
- The system shall allow students to view bus route information.
- The system shall allow students to view bus schedules.
- The system shall allow students to track the current location of their assigned bus.
- The system shall allow students to view the estimated arrival time (ETA) of the bus.
- The system shall allow students to view alerts related to their assigned bus.

3. Driver Management

- The system shall store driver information including license number and address.
- The system shall allow drivers to log in and access their dashboard.
- The system shall allow drivers to view assigned bus details.
- The system shall allow drivers to update the current location of their assigned bus.
- The system shall allow drivers to update the ETA of their assigned bus.
- The system shall allow drivers to submit alerts regarding delays, route changes, emergencies, or breakdowns.

4. Bus Management

- The system shall maintain records of buses including registration number, capacity, and model.
- The system shall associate each bus with a route.
- The system shall allow administrators to add, update, and delete bus records.
- The system shall allow administrators to view all buses.

5. Route Management

- The system shall maintain route information including starting and ending points.

- The system shall allow administrators to create routes.
- The system shall allow administrators to modify route information.
- The system shall allow administrators to delete routes.
- The system shall associate buses with routes.

6. Stop Management

- The system shall maintain information about route stops.
- The system shall allow administrators to add stops to routes.
- The system shall allow administrators to modify stop information.
- The system shall allow administrators to delete stops.
- The system shall display all stops for a selected route.

7. Bus Assignment Management

- The system shall assign drivers to buses.
- The system shall maintain assignment records and assignment end dates.
- The system shall allow administrators to update driver-bus assignments.
- The system shall allow administrators to view all assignments.

8. Schedule Management

- The system shall maintain bus departure and arrival schedules.
- The system shall allow administrators to create schedules.
- The system shall allow administrators to modify schedules.
- The system shall allow administrators to delete schedules.
- The system shall allow students and drivers to view schedules.

9. Bus Tracking

- The system shall store the current latitude and longitude of buses.
- The system shall record the timestamp of each location update.
- The system shall maintain ETA information for each bus.
- The system shall display real-time bus location information.
- The system shall display the latest location update timestamp.

10. Alert Management

- The system shall allow drivers to create alerts.
- The system shall store alert messages and timestamps.
- The system shall allow students to view alerts related to their assigned buses.
- The system shall allow administrators to view all alerts.

11. Administration

- The system shall allow administrators to manage users.
- The system shall allow administrators to manage buses, routes, stops, schedules, and assignments.
- The system shall allow administrators to view tracking information of all buses.
- The system shall allow administrators to monitor alerts generated by drivers.

2. Database Normalization

Database normalization is the process of organizing data into tables to eliminate redundancy (repetition) and prevent data inconsistencies during updates, insertions, or deletions.

Unnormalized Form (UNF)

In Easy Words: *Data is kept in one massive single list where information is repeated continuously, and column fields contain multiple blended values.*

CBTA_ALL_DATA Schema:

(USER_ID, NAME, EMAIL, PHONE_NUMBER, USERNAME, PASSWORD, ROLE, REGISTRATION_NO, DEPARTMENT, SEMESTER, EMPLOYEE_NO, DESIGNATION, LICENSE_NO, ADDRESS, BUS_ID, BUS_REGISTRATION_NO, CAPACITY, MODEL, ROUTE_ID, START_POINT, END_POINT, STOP_ID, STOP_NAME, LOCATION, ASSIGNMENT_ID, END_DATE, DEPARTURE_TIME, ARRIVAL_TIME, SCHEDULE_ID, ALERT_ID, MESSAGE, TIMESTAMP, LOCATION_ID, LATITUDE, LONGITUDE, ETA)

First Normal Form (1NF)

Every row-column cell must contain only a single atomic value, and all hidden commas or row groups are expanded into separated entries.

To convert UNF to 1NF, all multi-valued attributes are broken up. Due to combining everything into one structure, a wide composite primary key handles rows uniquely.

Composite Primary Key: (USER_ID, BUS_ID, STOP_ID, ASSIGNMENT_ID, SCHEDULE_ID, ALERT_ID, LOCATION_ID)

Second Normal Form (2NF)

The database must be in 1NF, and tables are separated so that attributes depend only on their full corresponding unique identifier key.

We split the giant table into separate tables based on key dependency clusters, completely removing partial dependencies. Additionally, following standard relational design rules, the pure

many-to-many relationship table STUDENT_BUS is defined by its precise composite primary key (STUDENT_ID, BUS_ID).

Separated 2NF Tables Structure:

Table: USERS

Columns: USER_ID (PK), NAME, EMAIL, PHONE_NUMBER, USERNAME, PASSWORD, ROLE

Table: STUDENT

Columns: USER_ID (PK/FK), REGISTRATION_NO, DEPARTMENT, SEMESTER

Table: DRIVER

Columns: USER_ID (PK/FK), LICENSE_NO, ADDRESS

Table: ADMIN

Columns: USER_ID (PK/FK), EMPLOYEE_NO, DESIGNATION

Table: BUS

Columns: BUS_ID (PK), BUS_REGISTRATION_NO, CAPACITY, MODEL, ROUTE_ID, START_POINT, END_POINT

Table: STOP

Columns: STOP_ID (PK), STOP_NAME, LOCATION, ROUTE_ID

Table: ASSIGNMENT**Columns:** ASSIGNMENT_ID (PK), DRIVER_ID, BUS_ID, END_DATE**Table: SCHEDULE****Columns:** SCHEDULE_ID (PK), DEPARTURE_TIME, ARRIVAL_TIME, BUS_ID, ROUTE_ID**Table: ALERT****Columns:** ALERT_ID (PK), MESSAGE, TIMESTAMP, BUS_ID**Table: LOCATION****Columns:** LOCATION_ID (PK), LATITUDE, LONGITUDE, TIMESTAMP, ETA, BUS_ID**Table: STUDENT_BUS****Columns:** STUDENT_ID (PK/FK), BUS_ID (PK/FK) [Composite Primary Key]**Third Normal Form (3NF)**

The database must be in 2NF, and no non-key field should depend indirectly on the primary key through another non-key field.

In 2NF, the BUS table contained a transitive chain: BUS_ID -> ROUTE_ID -> START_POINT, END_POINT. To clear this, path fields are extracted into an isolated ROUTE table.

Final Separated 3NF Normalized Database Tables:

Table: USERS

Columns: USER_ID (PK), NAME, EMAIL, PHONE_NUMBER, USERNAME, PASSWORD, ROLE

Table: STUDENT

Columns: USER_ID (PK/FK to USERS), REGISTRATION_NO, DEPARTMENT, SEMESTER

Table: DRIVER

Columns: USER_ID (PK/FK to USERS), LICENSE_NO, ADDRESS

Table: ADMIN

Columns: USER_ID (PK/FK to USERS), EMPLOYEE_NO, DESIGNATION

Table: BUS (Modified to 3NF)

Columns: BUS_ID (PK), BUS_REGISTRATION_NO, CAPACITY, MODEL, ROUTE_ID (FK to ROUTE)

Table: ROUTE (Newly Isolated in 3NF)

Columns: ROUTE_ID (PK), START_POINT, END_POINT

Table: STOP

Columns: STOP_ID (PK), STOP_NAME, LOCATION, ROUTE_ID (FK to ROUTE)

Table: ASSIGNMENT

Columns: ASSIGNMENT_ID (PK), DRIVER_ID (FK to DRIVER), BUS_ID (FK to BUS), END_DATE

Table: SCHEDULE

Columns: SCHEDULE_ID (PK), DEPARTURE_TIME, ARRIVAL_TIME, BUS_ID (FK to BUS), ROUTE_ID (FK to ROUTE)

Table: ALERT

Columns: ALERT_ID (PK), MESSAGE, TIMESTAMP, BUS_ID (FK to BUS)

Table: LOCATION

Columns: LOCATION_ID (PK), LATITUDE, LONGITUDE, TIMESTAMP, ETA, BUS_ID (FK to BUS)

Table: STUDENT_BUS

Columns: STUDENT_ID (PK/FK to STUDENT), BUS_ID (PK/FK to BUS) [Composite Primary Key]

3. SQL Database Schema & Implementation Scripts

```
CREATE DATABASE CBTA;
USE CBTA;

CREATE TABLE USERS(
    USER_ID INT PRIMARY KEY AUTO_INCREMENT,
    NAME VARCHAR(255),
    EMAIL VARCHAR(255) UNIQUE,
    PHONE_NUMBER VARCHAR(15),
    USERNAME VARCHAR(255) UNIQUE,
    PASSWORD VARCHAR(255),
    ROLE ENUM('STUDENT', 'ADMIN', 'DRIVER') NOT NULL
);

CREATE TABLE STUDENT(
    USER_ID INT PRIMARY KEY,
    REGISTRATION_NO INT UNIQUE,
    DEPARTMENT VARCHAR(255),
    SEMESTER VARCHAR(255),
    FOREIGN KEY(USER_ID) REFERENCES USERS(USER_ID)
);

CREATE TABLE ADMIN(
    USER_ID INT PRIMARY KEY,
    EMPLOYEE_NO VARCHAR(20) UNIQUE,
    DESIGNATION VARCHAR(100),
    FOREIGN KEY(USER_ID) REFERENCES USERS(USER_ID)
);

CREATE TABLE DRIVER(
    USER_ID INT PRIMARY KEY,
    LICENSE_NO VARCHAR(255) UNIQUE,
    ADDRESS VARCHAR(255),
    FOREIGN KEY(USER_ID) REFERENCES USERS(USER_ID)
);

CREATE TABLE ROUTE(
    ROUTE_ID INT PRIMARY KEY,
    START_POINT VARCHAR(255),
    END_POINT VARCHAR(255)
);

CREATE TABLE BUS(
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        BUS_ID INT PRIMARY KEY,
        REGISTRATION_NO VARCHAR(255) UNIQUE,
        CAPACITY INT,
        MODEL VARCHAR(255),
        ROUTE_ID INT,
        FOREIGN KEY(ROUTE_ID) REFERENCES ROUTE(ROUTE_ID)
    );

CREATE TABLE STUDENT_BUS(
    STUDENT_ID INT,
    BUS_ID INT,
    PRIMARY KEY(STUDENT_ID, BUS_ID),
    FOREIGN KEY(STUDENT_ID) REFERENCES STUDENT(USER_ID),
    FOREIGN KEY(BUS_ID) REFERENCES BUS(BUS_ID)
);

CREATE TABLE STOP(
    STOP_ID INT PRIMARY KEY,
    STOP_NAME VARCHAR(255),
    LOCATION VARCHAR(255),
    ROUTE_ID INT,
    FOREIGN KEY(ROUTE_ID) REFERENCES ROUTE(ROUTE_ID)
);

CREATE TABLE ASSIGNMENT(
    ASSIGNMENT_ID INT PRIMARY KEY,
    DRIVER_ID INT,
    BUS_ID INT,
    END_DATE DATE,
    FOREIGN KEY(BUS_ID) REFERENCES BUS(BUS_ID),
    FOREIGN KEY(DRIVER_ID) REFERENCES DRIVER(USER_ID)
);

CREATE TABLE SCHEDULE(
    SCHEDULE_ID INT PRIMARY KEY,
    DEPARTURE_TIME TIME,
    ARRIVAL_TIME TIME,
    BUS_ID INT,
    ROUTE_ID INT,
    FOREIGN KEY(ROUTE_ID) REFERENCES ROUTE(ROUTE_ID),
    FOREIGN KEY(BUS_ID) REFERENCES BUS(BUS_ID)
);

CREATE TABLE ALERT(
    ALERT_ID INT PRIMARY KEY,
    MESSAGE VARCHAR(255),
    TIMESTAMP DATETIME,
    BUS_ID INT,
    FOREIGN KEY(BUS_ID) REFERENCES BUS(BUS_ID)
);

CREATE TABLE LOCATION(
    LOCATION_ID INT PRIMARY KEY,
    LATITUDE DECIMAL(10, 7),
    LONGITUDE DECIMAL(10, 7),

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        TIMESTAMP DATETIME,
        ETA INT,
        BUS_ID INT,
        FOREIGN KEY (BUS_ID) REFERENCES BUS (BUS_ID)
    );

INSERT INTO USERS (NAME, EMAIL, PHONE_NUMBER, USERNAME, PASSWORD, ROLE) VALUES
('Ali Khan', 'ali@gmail.com', '03001234567', 'ali123', 'pass123', 'STUDENT'),
('Ahmed Raza', 'raza@gmail.com', '03007654321', 'raza123', 'pass123', 'DRIVER'),
('Sara Ali', 'sara@gmail.com', '03111234567', 'sara123', 'pass123', 'ADMIN'),
('Noor Fatima', 'noor@gmail.com', '03007654321', 'noor123', 'pass123', 'STUDENT'),
('Iqbal', 'Iqbal@gmail.com', '03003456789', 'kamran123', 'pass123', 'DRIVER'),
('Sana Malik', 'sana@gmail.com', '03115678901', 'sana123', 'pass123', 'STUDENT'),
('Hira', 'hira@gmail.com', '03004567890', 'hira123', 'pass123', 'STUDENT'),
('Usman', 'usman@gmail.com', '03006789012', 'usman123', 'pass123', 'STUDENT'),
('Ahmed', 'Ahmed@gmail.com', '03112345678', 'Ahmed123', 'pass123', 'DRIVER'),
('Bilal', 'Bilal@gmail.com', '03009876543', 'Bilal123', 'pass123', 'DRIVER');

INSERT INTO STUDENT (USER_ID, REGISTRATION_NO, DEPARTMENT, SEMESTER) VALUES
(1, 101, 'Computer Science', '5th'),
(4, 102, 'Zoology', '5th'),
(6, 105, 'Zoology', '3rd'),
(7, 103, 'Chemistry', '5th'),
(8, 104, 'Computer Science', '5th');

INSERT INTO DRIVER (USER_ID, LICENSE_NO, ADDRESS) VALUES
(2, 'LIC12345', 'Gulberg'),
(5, 'LIC23456', 'Bahria Town'),
(9, 'LIC34567', 'Satellite Town'),
(10, 'LIC45678', 'Eagle City');

INSERT INTO ADMIN (USER_ID, EMPLOYEE_NO, DESIGNATION) VALUES
(3, 'EMP101', 'Transport Manager');

INSERT INTO ROUTE (ROUTE_ID, START_POINT, END_POINT) VALUES
(1, 'University Campus, Sargodha', 'Kot Momin'),
(2, 'University Campus, Sargodha', 'Bhalwal'),
(3, 'University Campus, Sargodha', 'Bhera'),
(4, 'University Campus, Sargodha', 'Sillanwali'),
(5, 'University Campus, Sargodha', 'Rawalpindi');

INSERT INTO BUS (BUS_ID, REGISTRATION_NO, CAPACITY, MODEL, ROUTE_ID) VALUES
(1, 'LEA-1234', 50, 'Hino Bus', 1),
(2, 'ABC-5678', 50, 'Hino Bus', 2),
(3, 'XYZ-9012', 50, 'Hino Bus', 3),
(4, 'PQR-3456', 50, 'Hino Bus', 4),
(5, 'LMN-7890', 50, 'Hino Bus', 5);

INSERT INTO STUDENT_BUS (STUDENT_ID, BUS_ID) VALUES
(1, 1), (4, 1), (7, 2), (8, 3);

INSERT INTO STOP (STOP_ID, STOP_NAME, LOCATION, ROUTE_ID) VALUES
(1, 'University Gate', 'Sargodha', 1),
(2, 'Kot Momin Stop', 'Kot Momin', 1),
(3, 'Bhalwal Road Stop', 'Sargodha Road', 2),

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(4, 'Bhalwal Stop', 'Bhalwal', 2),
(5, 'Bhera Road Stop', 'Bhera Road', 3),
(6, 'Bhera Stop', 'Bhera', 3),
(7, 'Sillanwali Road Stop', 'Sargodha Road', 4),
(8, 'Sillanwali Stop', 'Sillanwali', 4),
(9, 'Motorway Stop', 'M2 Interchange', 5),
(10, 'Rawalpindi Stop', 'Rawalpindi', 5);

INSERT INTO ASSIGNMENT (ASSIGNMENT_ID, DRIVER_ID, BUS_ID, END_DATE) VALUES
(1, 2, 1, '2026-12-31'),
(2, 5, 2, '2026-12-31'),
(3, 9, 3, '2026-12-31'),
(4, 10, 4, '2026-12-31'),
(5, 2, 5, '2026-12-31');

INSERT INTO SCHEDULE (SCHEDULE_ID, DEPARTURE_TIME, ARRIVAL_TIME, BUS_ID, ROUTE_ID)
VALUES
(1, '08:00:00', '09:00:00', 1, 1),
(2, '10:00:00', '11:30:00', 2, 2),
(3, '07:30:00', '08:30:00', 3, 3),
(4, '09:00:00', '10:00:00', 4, 4),
(5, '11:00:00', '12:00:00', 5, 5);

INSERT INTO ALERT (ALERT_ID, MESSAGE, TIMESTAMP, BUS_ID) VALUES
(1, 'Bus delayed due to traffic', '2026-03-26 08:15:00', 1),
(2, 'Breakdown reported', '2026-03-26 10:05:00', 2),
(3, 'Route changed', '2026-03-26 07:50:00', 3),
(4, 'Emergency stop', '2026-03-26 09:20:00', 4),
(5, 'Minor delay due to weather', '2026-03-26 11:15:00', 5);

INSERT INTO LOCATION (LOCATION_ID, LATITUDE, LONGITUDE, TIMESTAMP, ETA, BUS_ID)
VALUES
(1, 33.684422, 73.047882, '2026-03-26 08:30:00', 5, 1),
(2, 33.565110, 73.016913, '2026-03-26 10:15:00', 8, 2),
(3, 33.678900, 73.055000, '2026-03-26 07:45:00', 10, 3),
(4, 33.610000, 72.980000, '2026-03-26 09:05:00', 6, 4),
(5, 33.700000, 73.100000, '2026-03-26 11:20:00', 12, 5);

COMMIT;
```

4. EER Diagram :

